

ANÁLISIS DE FACTORES ACADÉMICOS Y SU RELACIÓN CON LA DEPRESIÓN ESTUDIANTIL

Inteligencia artificial I E2

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OBJETIVO

Clasificar estudiantes si tiene o no depresión dependiendo de distintos factores académicos y personales.

DATASET

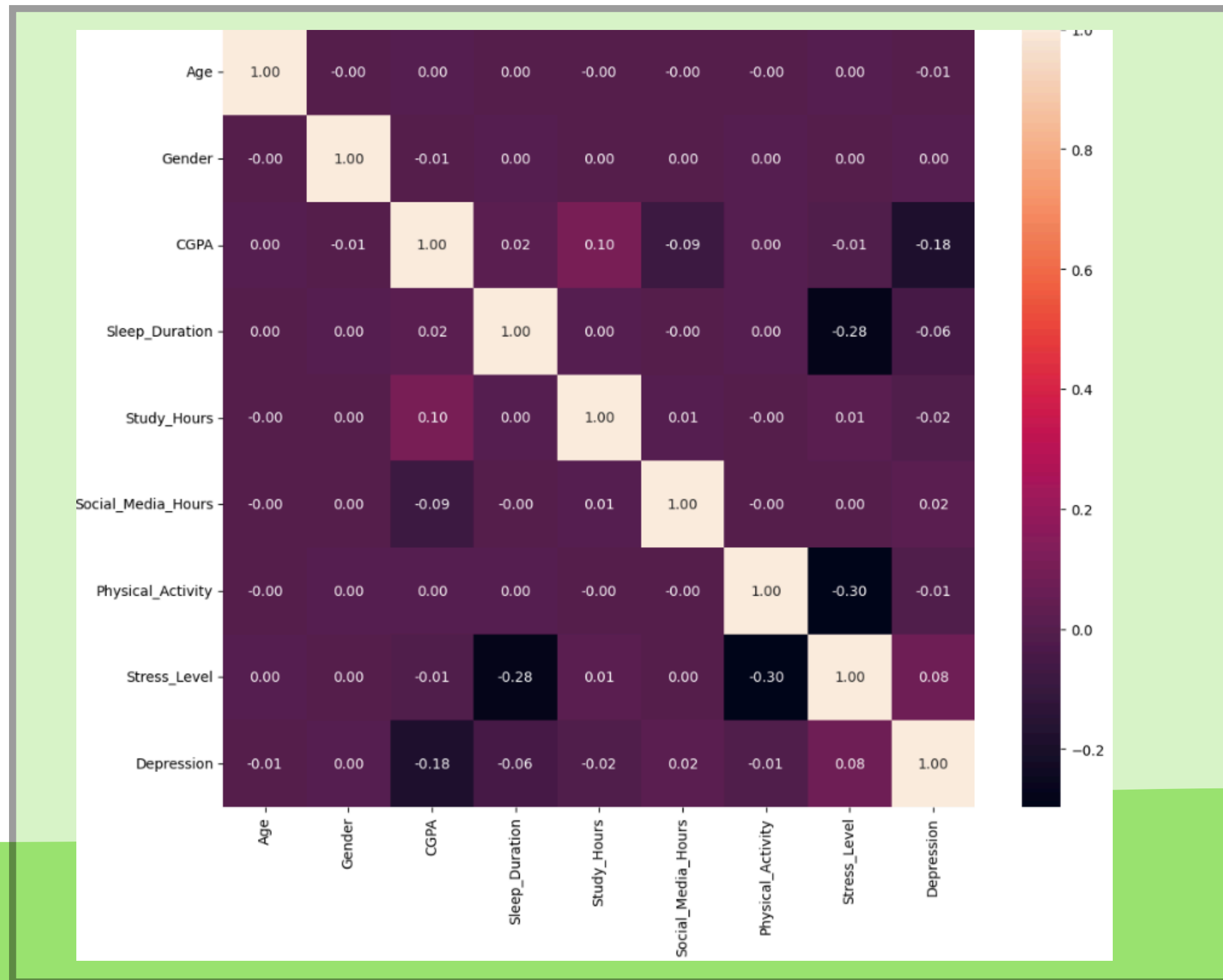
<https://www.kaggle.com/datasets/aldinwhyudii/student-depression-and-lifestyle-100k-data>

Contiene 10.000 filas y 11 columnas, con un porcentaje del 10% de estudiantes con depresión

DATASET

Student_ID	Unique identifier for each student.	Integer
Age	Age of the student (18-24).	Integer
Gender	Gender of the student (Male/Female).	String
Department	Field of study (Engineering, Business, Arts, etc.).	String
CGPA	Cumulative Grade Point Average (0.0 - 4.0).	Float
Sleep_Duration	Average hours of sleep per night.	Float
Study_Hours	Average hours spent studying per day.	Float
Social_Media_Hours	Average hours spent on social media per day.	Float
Physical_Activity	Average minutes of physical activity per week.	Integer
Stress_Level	Self-reported stress level (scale 0-10).	Integer
Depression	Mental health status (True = Probable Depression, False = Healthy).	Boolean

CORRELACIÓN

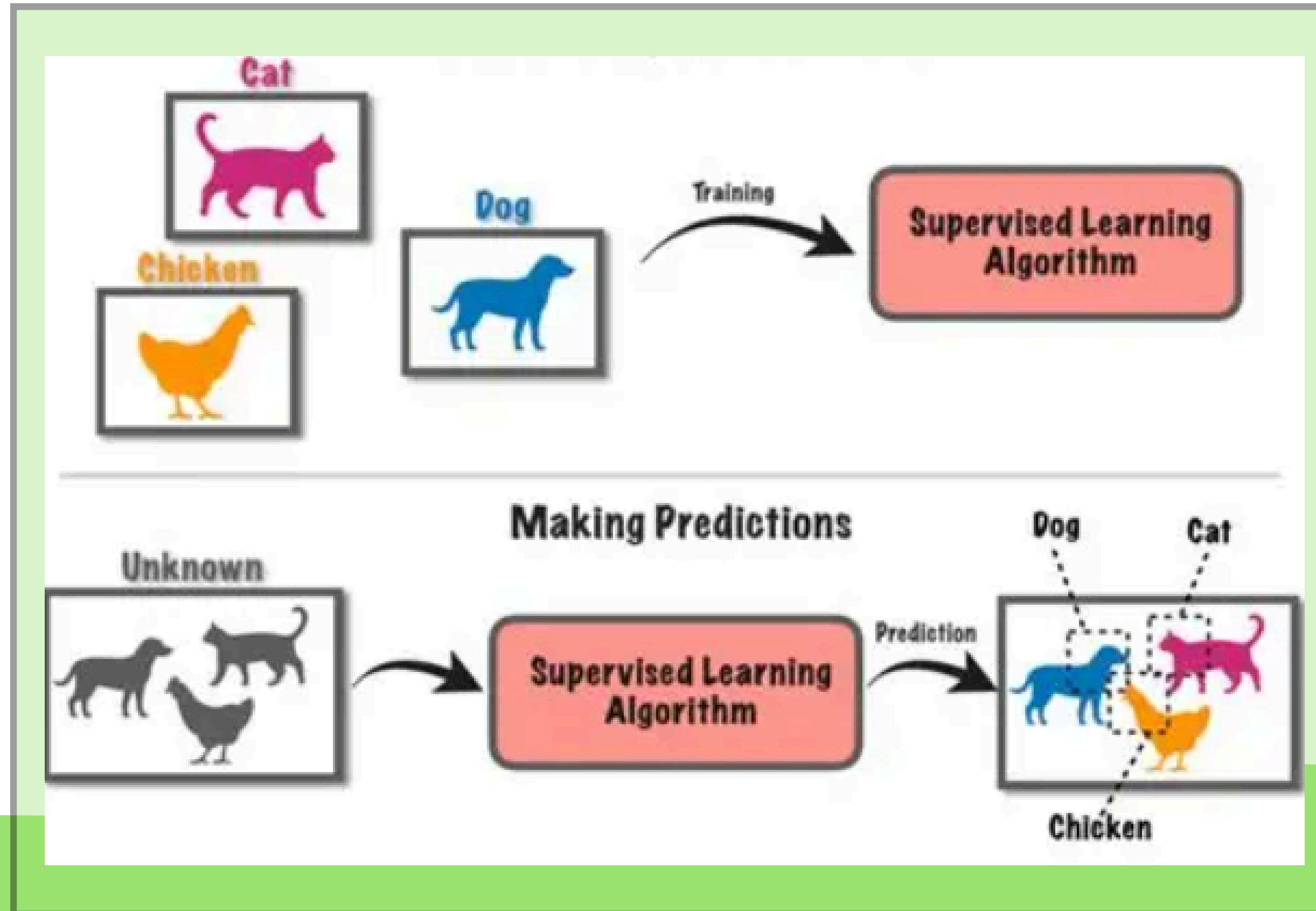


PRE-PROCESAMIENTO

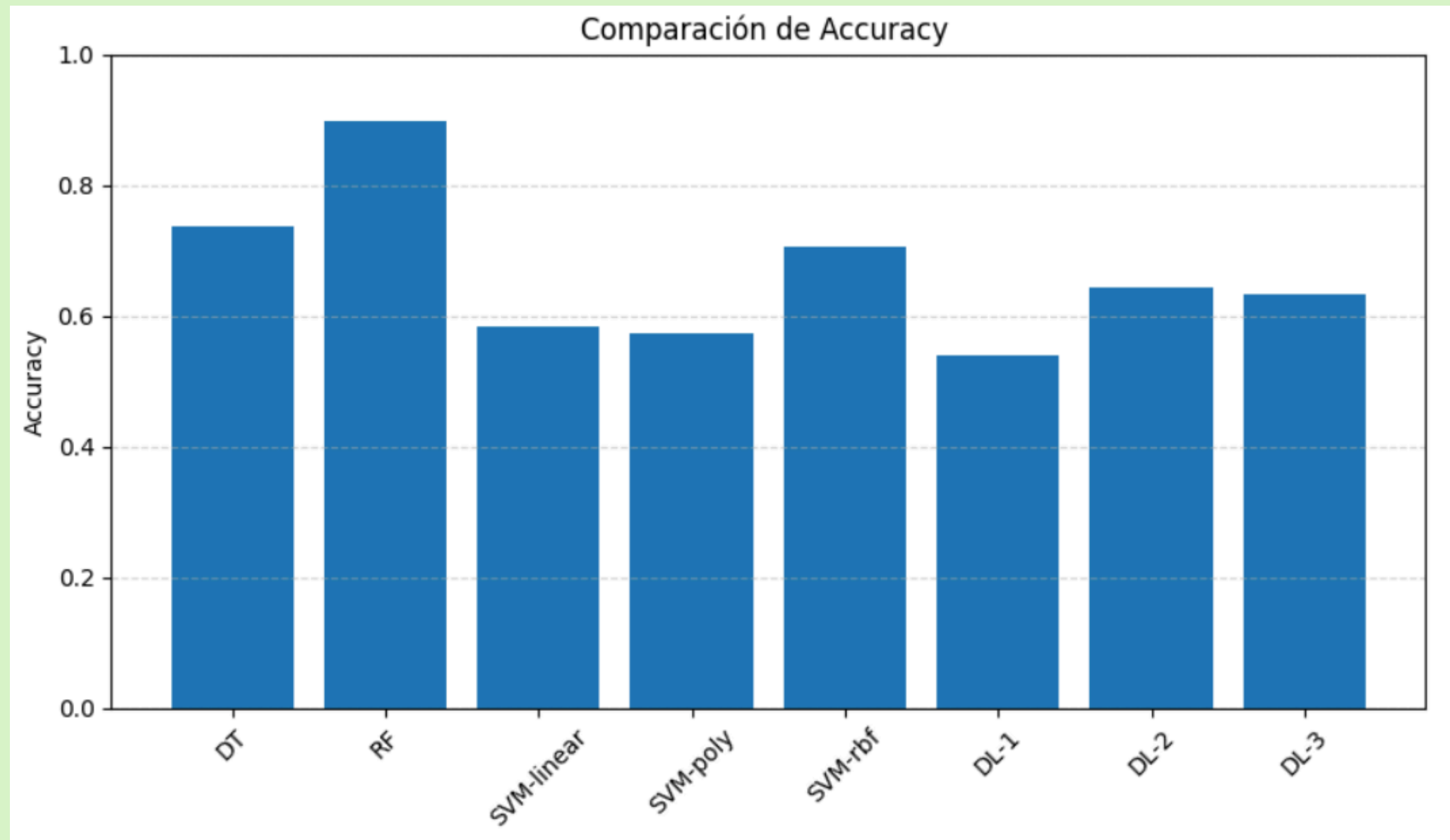
```
df = df.drop(columns=["Student_ID"])
df = df.drop(columns=["Department"])
df["Gender"] = df["Gender"].map({"Male": 0, "Female": 1})

df["Depression"] = df["Depression"].map({False: 0, True: 1})
print(df)
```

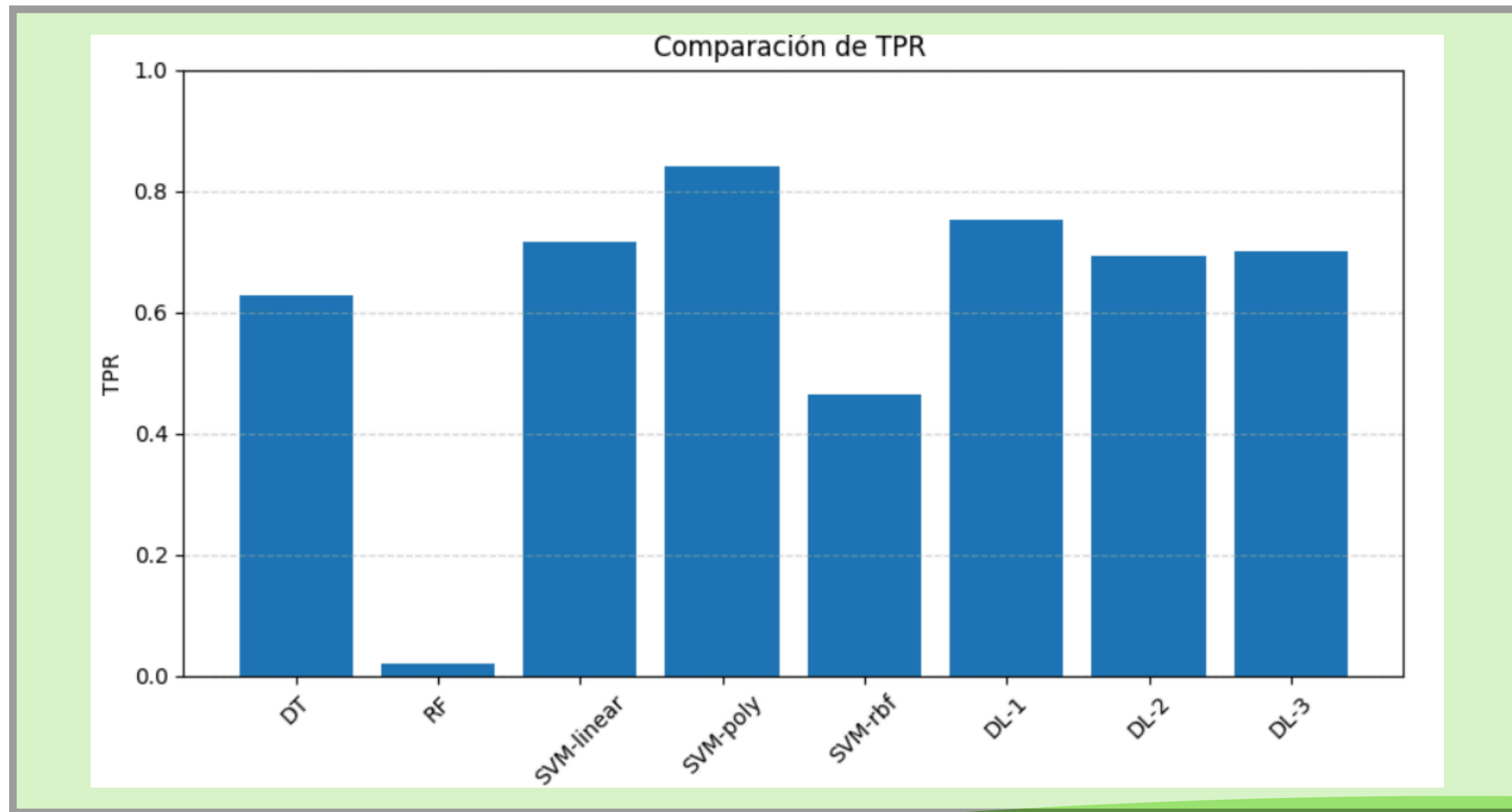
APRENDIZAJE SUPERVISADO



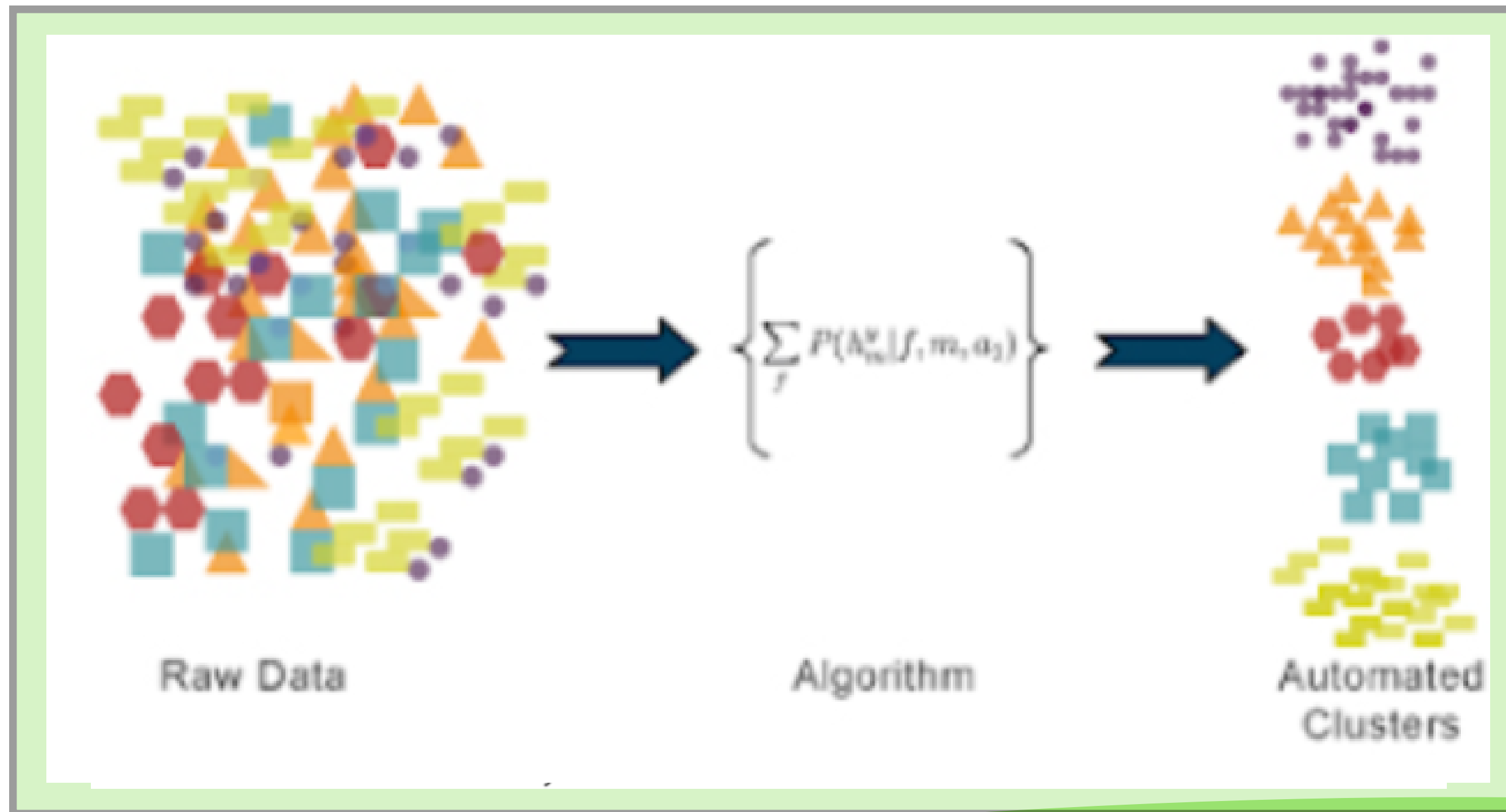
Accuracy



TPR



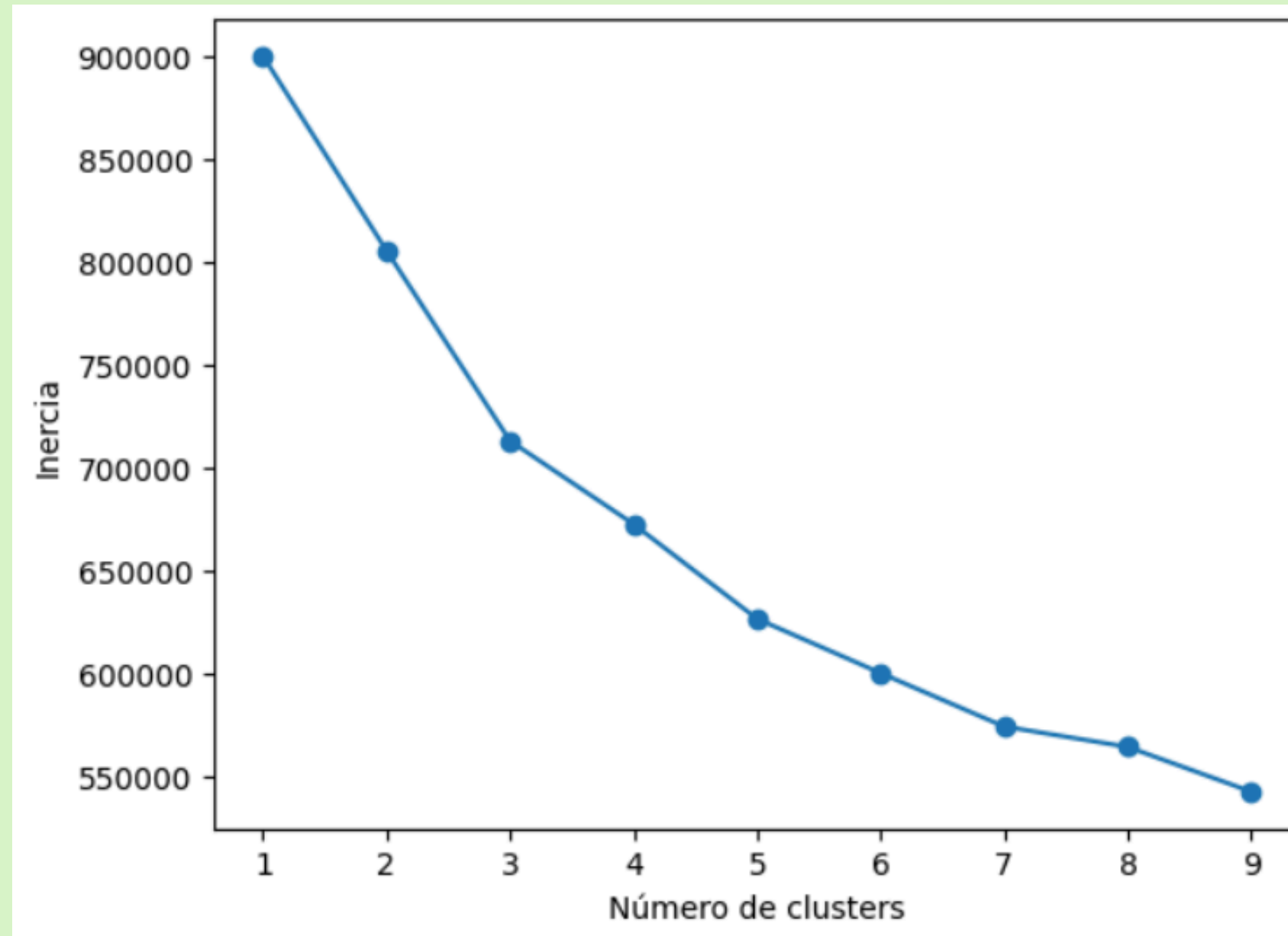
APRENDIZAJE NO SUPERVISADO



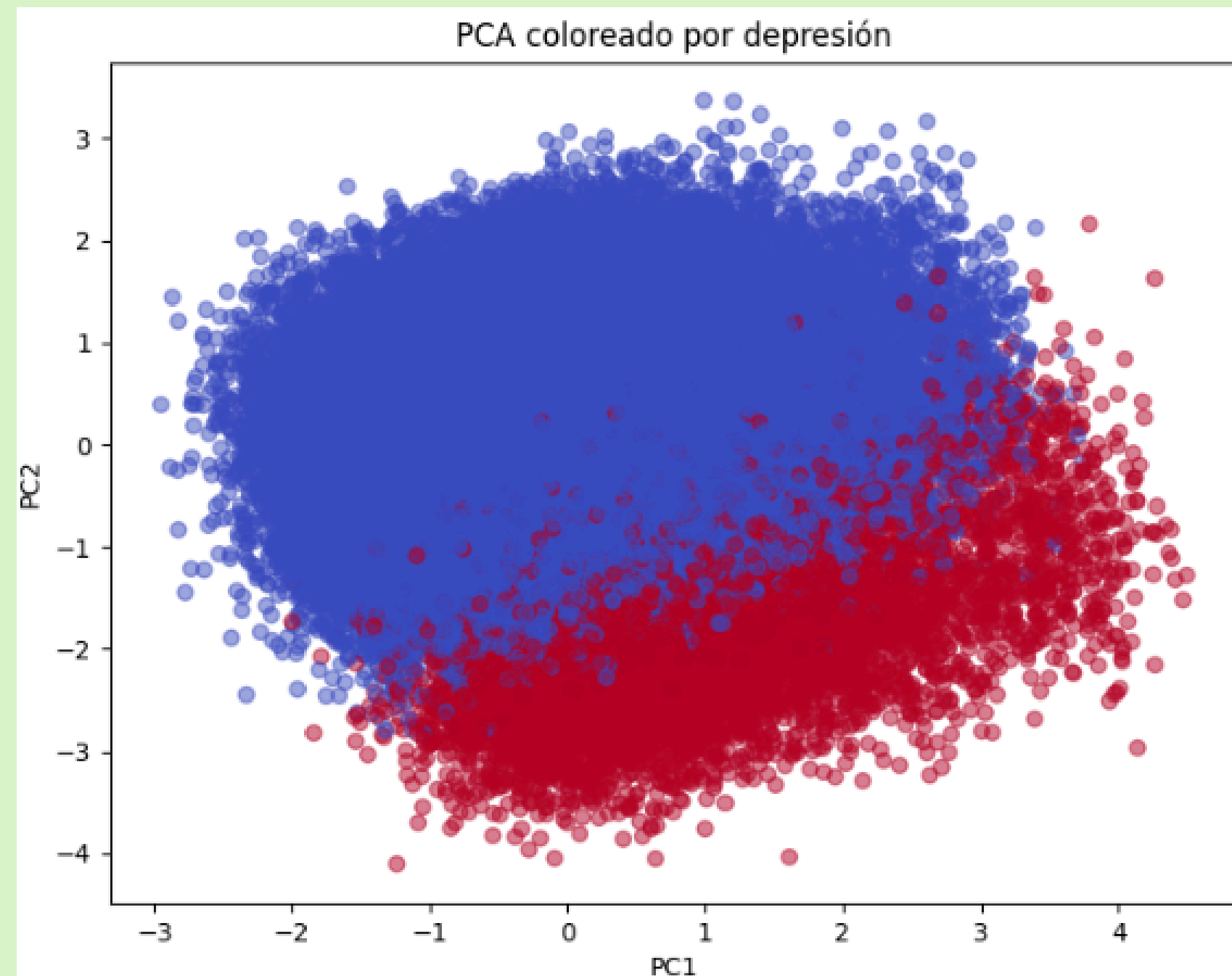
StandardScaler

```
scaler = StandardScaler()  
scaled_df = scaler.fit_transform(df)  
sclar_df = pd.DataFrame(scaled_df, columns=df.columns)  
print(sclar_df)
```

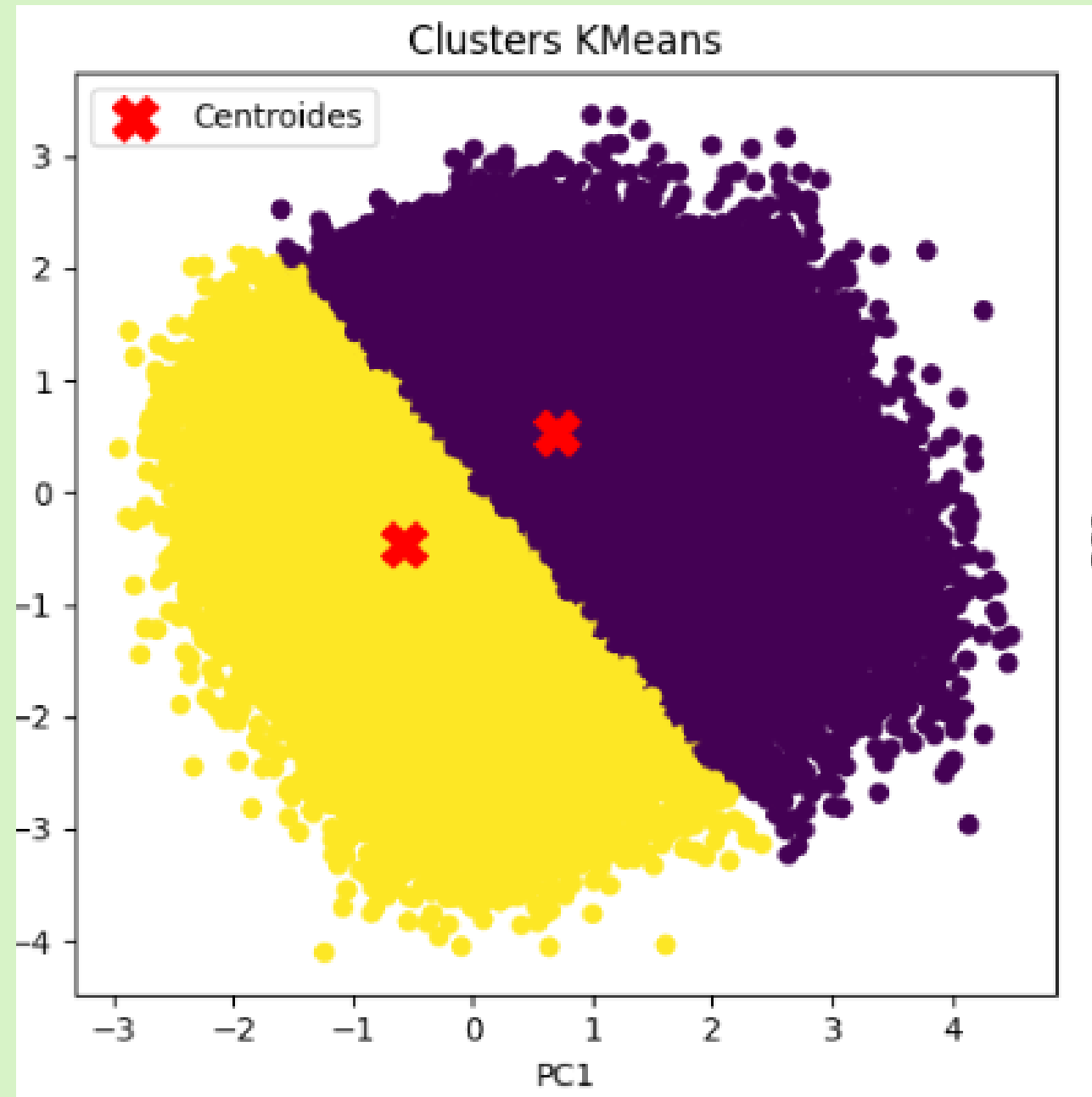
Número de clusters con el método de



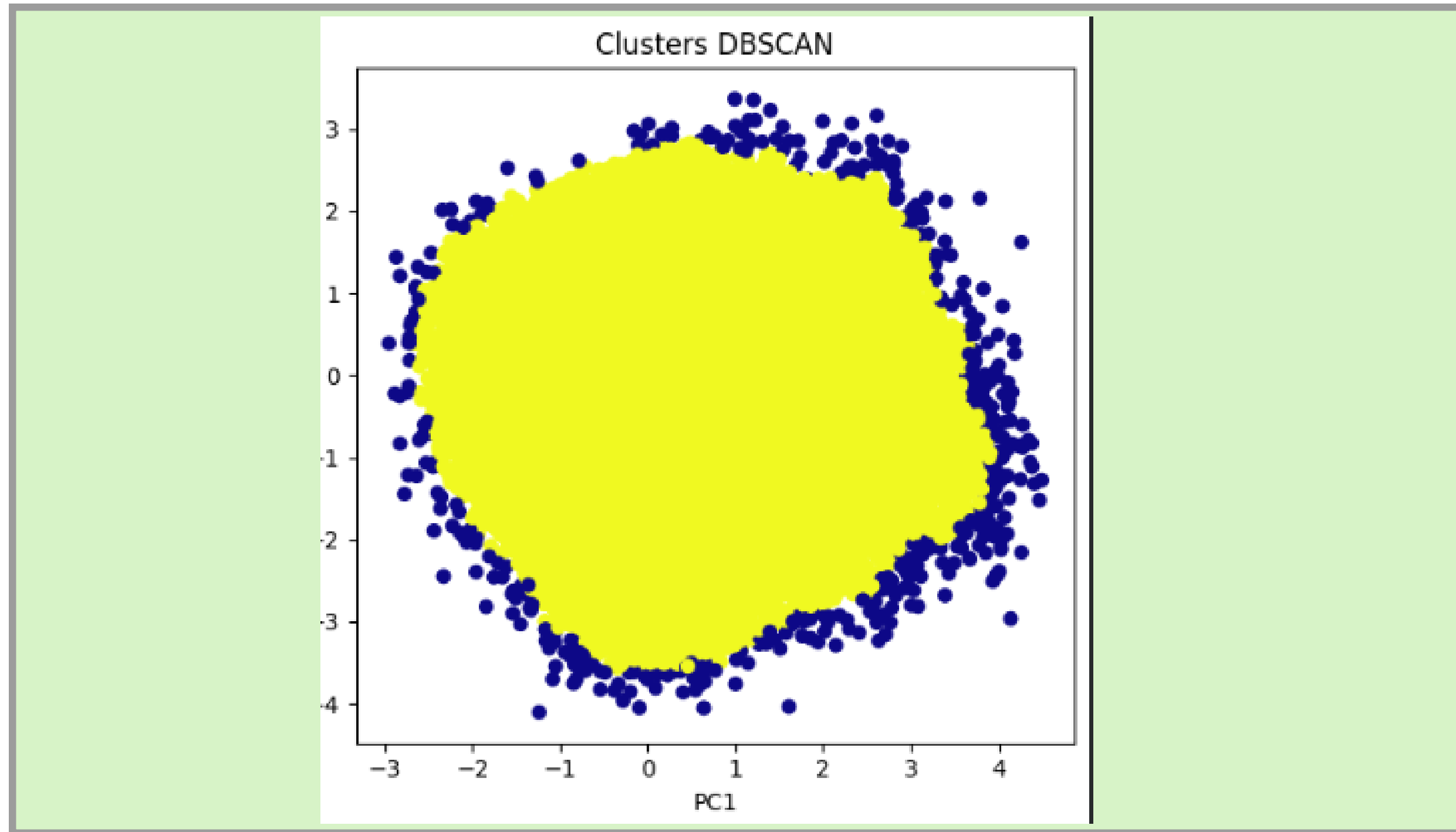
PCA



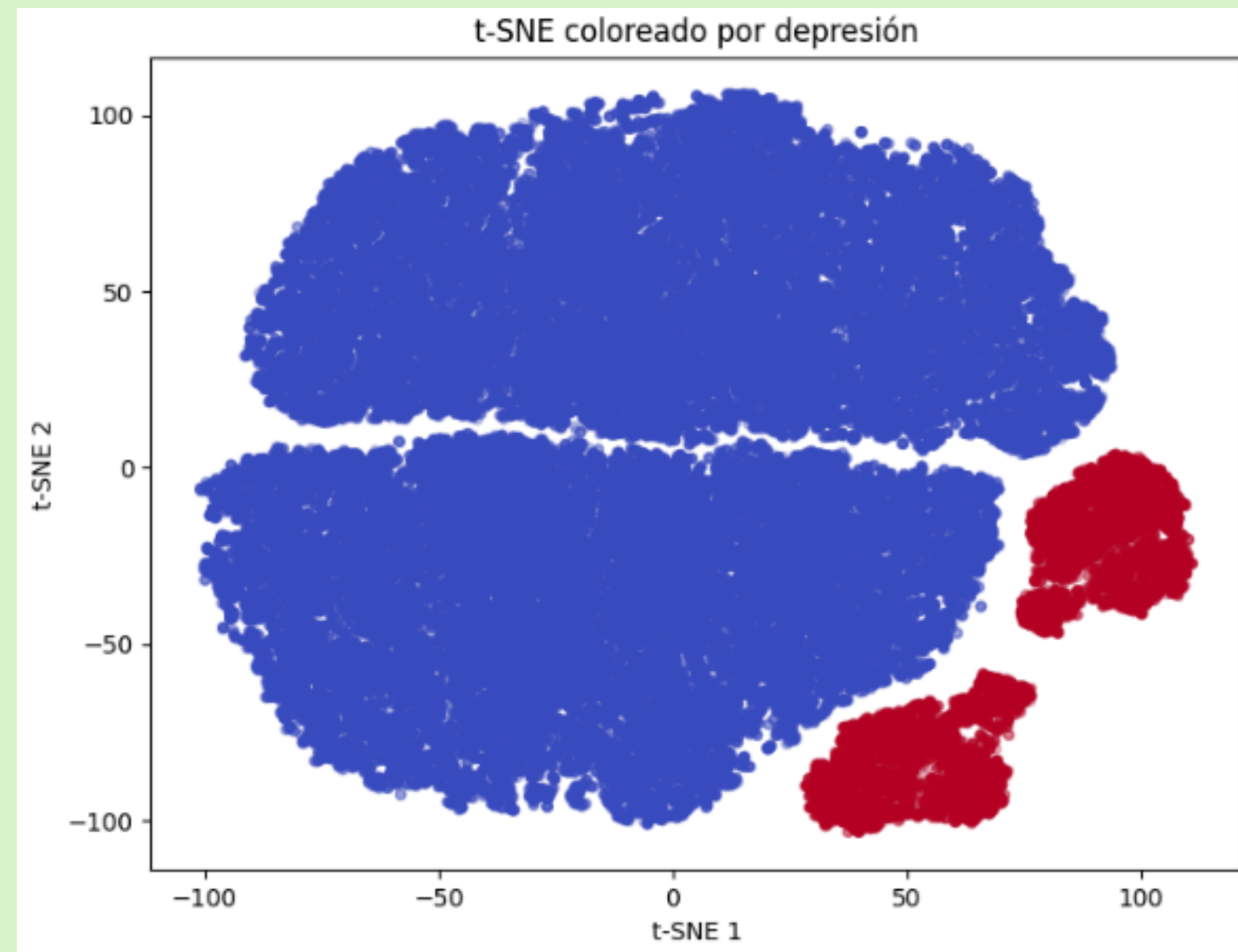
KMEANS



DBSCAN



T-SNE



MUCHAS GRACIAS